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Paleomagnetism of a Polarity Transition in the Lower (?) Triassic Chugwater Formation, Wyoming

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A total of 156 oriented specimens, taken at stratigraphic intervals of 3 to 20 cm, have been collected from the Red Peak Member of the Chugwater Formation at three localities approximately 25 km southeast of Dubois, Wyoming. The stratigraphic interval containing the polarity transition in the remanent magnetic field is about 1.0 m. All specimens were thermally demagnetized at 500°, 550°, 620°, and 660°C and show directions of magnetization of normal, reversed, and transitional character as well as a remarkable decrease of the remanent intensity in the transition zone. The section sampled is divided into five intensity periods: two periods recording the dipolar field, one recording a dipolar field of decreasing intensity, one recording a dipolar field of increasing intensity, and a period of low intensity recording the field present during the 180° directional swing. A virtual geomagnetic pole (VGP) path obtained for a N → R polarity transition makes a fairly smooth transit, with the path confined to a sector of longitude between 85°W and 110°E. The reversal's onset is first indicated by a slow movement of the VGP across northern latitudes. The VGP then lingers to two positions 60°N and 40°N (35°E to 70°E) and 20°N and 40°N (335°E to 5°E), suggesting brief periods of field stability (or nondipolar field dominance). The onset of the transition appears to be more rapid than the recovery. The VGP path observed at a northern hemisphere site is located about 90° from the site. The pole position derived from stable normal and reversed portions of this study (115.5°E, 46.4°N, α_{95} = 4.1, *dp* = 2.2, *dm* = 4.2) lies approximately in the center of the main group of poles reported for other Triassic units.

INTRODUCTION

One of the most interesting features of the earth's magnetic field is that worldwide simultaneous polarity changes have occurred. These polarity changes provide the most compelling evidence that the observations of reversed magnetization in rocks is related to reversals of the earth's magnetic field. Other evidence comes from measurements showing that the polarity changed continuously from one polarity to the other in a rock sequence, a polarity change or polarity transition [*McElhinny*, 1973]. Directions of magnetization observed in rocks during such changes are called intermediate or sometimes transitional and have been reported by several workers. They are not common, however, indicating that the duration of a transition is short [*Goldstein et al.*, 1969].

Most early estimates of the duration of the transition tended to give times of the order of 10^3 or 10^4 years. However, much longer transition times have been suggested [*Momose*, 1963]. *Cox and Dalrymple* [1967] estimated statistically that the time required for a reversal is 5000 years or less. Oceanic cores have yielded independent estimates of 4000 years [*Harrison and Somayajulu*, 1966] and 4600 years [*Opdyke et al.*, 1973], and oriented cores reported more recently by *Hammond et al.* [1979] suggested directional changes of approximately 6000 years for the Jaramillo and Olduvai events. The Boso Peninsula records give 4700 years [*Niitsuma*, 1971]. The common element linking these phenomena is that they occur very quickly, so quickly in fact that it is difficult to find rocks which have recorded them completely enough and accurately enough to provide useful magnetic records [*Cox et al.*, 1975].

As pointed out by *Cox et al.* [1975], a difficulty encountered in much of this research is that as magnetic recorders, many rocks have a poor high-frequency response. Volcanic rocks, for example, are erupted too sporadically to provide a record with a needed degree of continuity. Sedimentary rocks also contain many stratigraphic gaps corresponding to times of nondeposition or erosion. Moreover, an initially complete and accurate paleomagnetic record in sediments may be obliterated or (what is worse) altered by chemical diagenesis and bioturbation occurring after initial deposition [*Løvlie*, 1976; *Kent and Lowrie*, 1974]. The magnetic recorders on the earth have undergone erasure and overprinting and in general have generated records with numerous gaps. Furthermore, the paleomagnetic record from any single sediment core or geologic section is generally far from being complete or perfect. As a result, our picture of the fine structure of the ancient magnetic field is still hazy.

The present work was undertaken in spite of the above enumerated difficulties to determine whether or not the paleomagnetism of red sediments could provide additional insight into the nature of the geomagnetic field during periods of transition between normal and reversed polarity.

GEOLOGY

The Chugwater Formation (Triassic) is well exposed in central Wyoming where it is a thick, conspicuous sequence of red beds exposed around the flanks of most uplifts. Its regional stratigraphy and history are discussed by *Love* [1939], *McKee et al.* [1959], *Picard* [1967], and more recently by *High and Picard* [1969]. The nomenclature of the Chugwater Formation proposed by *Love* [1939] and followed by *McKee et al.* [1959] is used in this paper along with the modifications proposed by *High and Picard* [1969]. From oldest to youngest, Love's units of the Chugwater Formation are: Red Peak Member, Alcova Limestone Member, Crow Mountain Sandstone Member, and Popo Agie Member. *Picard* [1964] subdivides the Red Peak into five informal rock units; from oldest to youngest these are: silty claystone

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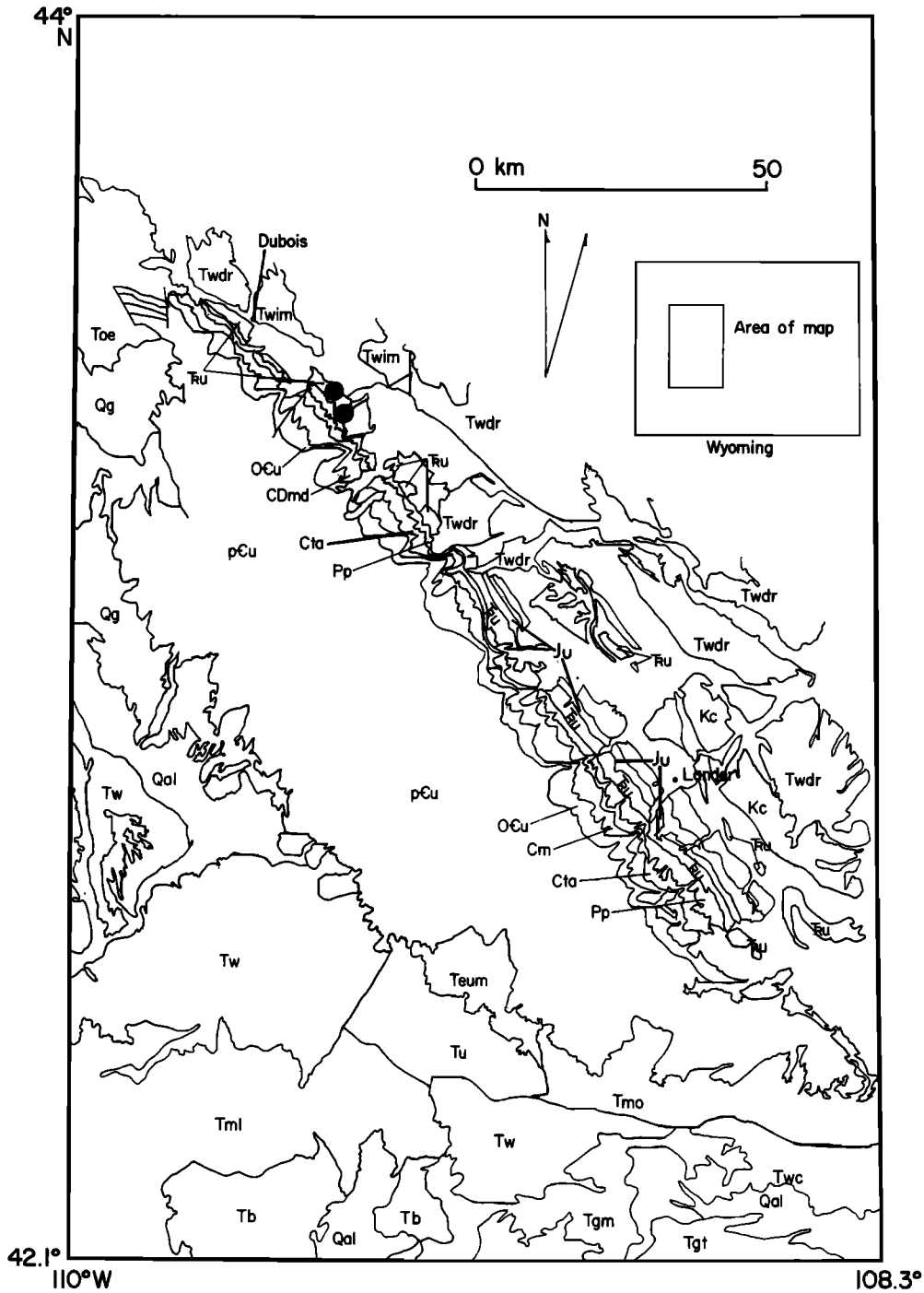


Fig. 1. Location map showing the sampling locality (solid circles) within the Chugwater Formation. Pp, Permian Phosphoria Formation; Tru, Triassic Chugwater Formation; Ju, Jurassic rocks, undivided (includes Nugget sandstone). Geology is from *Love et al.* [1955].

facies, lower platy facies, alternating facies, upper platy facies, and variegated facies.

The section sampled for these studies is in the alternating facies of the Red Peak Member located in west central Wyoming (Figure 1) southeast of the town of Dubois. Bedding within the Chugwater Formation dips at 9° to 15° in the region of study, and lithologic correlation between adjacent outcrops can be made easily. In the area of study, the Chugwater Formation overlies the Phosphoria Formation (Permian) and in the same area is overlain by the Nugget Sandstone (Jurassic?). Thus the age of the Chugwater Formation of this region is established as Triassic [*High and Picard*, 1969], although the formation itself contains few fossils. Despite the lack of abundant or diagnostic fossils, detailed stratigraphic relations have been established in large areas by means of physical correlations. On the basis of regional correlation with other Permian and Triassic strata, *High and Picard* [1969] conclude that the lowermost member of the Chugwater Formation may be of Early Triassic age. The uppermost units of the Chugwater Formation are con-

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sidered by the same authors to be as young as Late Triassic. Thus one can only say with fair certainty that the rocks studied are of Early Triassic age. No samples were taken from the portion of the section considered to be possibly of Middle or Upper Triassic corresponding to the Crow Mountain Member and the Popo Agie Member, respectively. All samples were collected below the Alcova Limestone, which separates the Red Peak Member and the Crow Mountain Member [*Love*, 1957; *High and Picard*, 1967, 1969; *Pipiringos*, 1968].

Sampling Procedure

Oriented samples were collected approximately 20 km southeast of Dubois, Wyoming (Figure 1). All samples were collected by means of a hand-held gasoline-powered drill [*Helsley*, 1967*b*] and were oriented by means of a magnetic compass.

The initial search for the reversal was done at a locality known as 'Red Grade,' 109°26.6'W, 43°24.8'N, on the south side of 'Little Red Creek' at elevations between 1975 and 1987 m where samples at widely spaced stratigraphic levels were drilled in order to identify the intervals of normal, reversed, and intermediate polarity. Succeeding samples at closer intervals focused on one reversal that occurred between two locally identifiable stratigraphic marker horizons (in this paper referred to as the upper clay seam and lower clay seam) that allowed stratigraphic correlations between sampling localities. Once the possible transition zone was stratigraphically identified, a total of 113 samples (approximately 156 specimens) were drilled at three localities: profile 1; profile 2, located 90 m west of profile 1; and profile 3, located 25 m west of profile 2 in the Red Peak Member of the Chugwater Formation. This sampling covered a stratigraphic interval of approximately 3.5 m of fine-grained red sediments and was drilled at 3- to 20-cm spacing between samples. The samples were drilled with inclinations ranging from 8° to 82° with respect to horizontal and were up to 25 cm in length. This procedure allows a more or less continuous sampling and in some cases provides an independent observation of the same stratigraphic level of the reversal. Since the Red Peak Member is about 200 to 300 m thick in west central Wyoming [*Picard*, 1964], the detailed results presented below cover only about one fiftieth of the total Chugwater section.

Samples collected during the initial search for reversals (of the order of 58 samples) were cut to 2.5-cm (1 inch) cylinders and measured in the field with a portable air turbine spinner magnetometer [*Helsley*, 1967*a*]. After the stratigraphic identification of the possible transition zone, subsequent samples taken in the continuous stratigraphic sequence were cut to 2.5-cm (1 inch) cylinders upon return to the laboratory and were initially measured on a PAR SM-1 magnetometer. Measurements made during the thermal demagnetization experiments, including remeasurement of the initial magnetization, were on a 6-cm vertical axis SCT cryogenic magnetometer.

Discussion of NRM Results

The natural remanent magnetization (NRM) of all samples is summarized in Figures 2 and 3. Several features are immediately apparent. The reversed samples have a declination near 150° and a slightly negative (upward) inclination. The normal sequence has a declination of about 340° and a positive inclination of about 23°. The initial grouping of the reversed and normal groups is remarkable.

Another conspicuous feature is what apparently looks like a polarity transition zone recorded from about 1.0 m to approximately 4.0 m for the three profiles (Figures 6 through 9). At this point this feature could represent either a true polarity transition or a superposition of normal and reverse components, or overprints of the present geomagnetic field, or a diagenetic feature such as a multiple antiparallel component acquired through chemical alteration over long time intervals. However, upon closer examination, none of these samples are near the field observed today and only one is near the axial dipolar field (Figure 2*a*).

Directions of both reversed and normal polarity are well grouped (Figures 2*a*–2*c* and 3*a*), and both groups have dispersions characteristic of many red bed sequences [e.g., *Helsley*, 1969]. The intermediate directions corresponding to the apparent polarity transition definitely do not show obvious evidence of streaking toward the present earth's magnetic field. The normal and reversed (NRM) groups give results that are very similar to other results from the Chugwater Formation reported by *Grubbs and Van der Voo* [1976] and *Van der Voo and Grubbs* [1977] and from other Lower Triassic units from the Colorado Plateau [*Irving*, 1964; *Helsley*, 1969; *Helsley and Steiner*, 1974; *Elston and Purucker*, 1979; *Purucker et al.*, 1980]. Thus both groups can be considered to be essentially stable and to represent a true direction for the Triassic period.

Demagnetization Experiments

Alternating field demagnetization of seven pilot samples was found to be ineffective in removing systematic spurious components even at field values of up to 280 mT. Similar results were found in red sediments of the Moenkopi Formation [*Helsley*, 1969]. This is probably due to a 'hard' viscous remanent magnetization (VRM) in hematite [*Dunlop and Stirling*, 1977]. Thus magnetic stability of the Chugwater samples was studied primarily by thermal methods.

Thermal demagnetization was accomplished in a conventional noninductively wound electrical furnace, the samples being cooled in a region where the residual field was less than 10 nT. Temperature was monitored in the furnace by means of a Chromel-Alumel thermocouple that was referred to an ice bath and read on a Beckman digital multimeter and a Love temperature controller.

Five pilot specimens were selected for a progressive thermal demagnetization experiment. These experiments produced changes in the direction of magnetization as well as effects on the observed intensity, until temperatures in excess of 550°C were reached. Figures 4, 5, and 6 show the response of the specimens to progressive heating experiments.

The normal specimens show a minor change in declination (7° to 14°) and inclination (6° up to 20°) (Figure 4*a*), and their intensities decrease by 30 to 40% at temperatures of less than 550° (see Figures 6 to 9). Transitional specimens, for example, specimen 126, show larger changes in declination (20°) and inclination (13°). The reversed sample shown in Figure 4 did not display systematic changes of more than 3° in declination and about 6° in inclination and there is little change in intensity at temperatures of less than 620°C (Figures 4*b* and 6). These results are comparable to the results from the Lower Triassic Moenkopi Formation sam-

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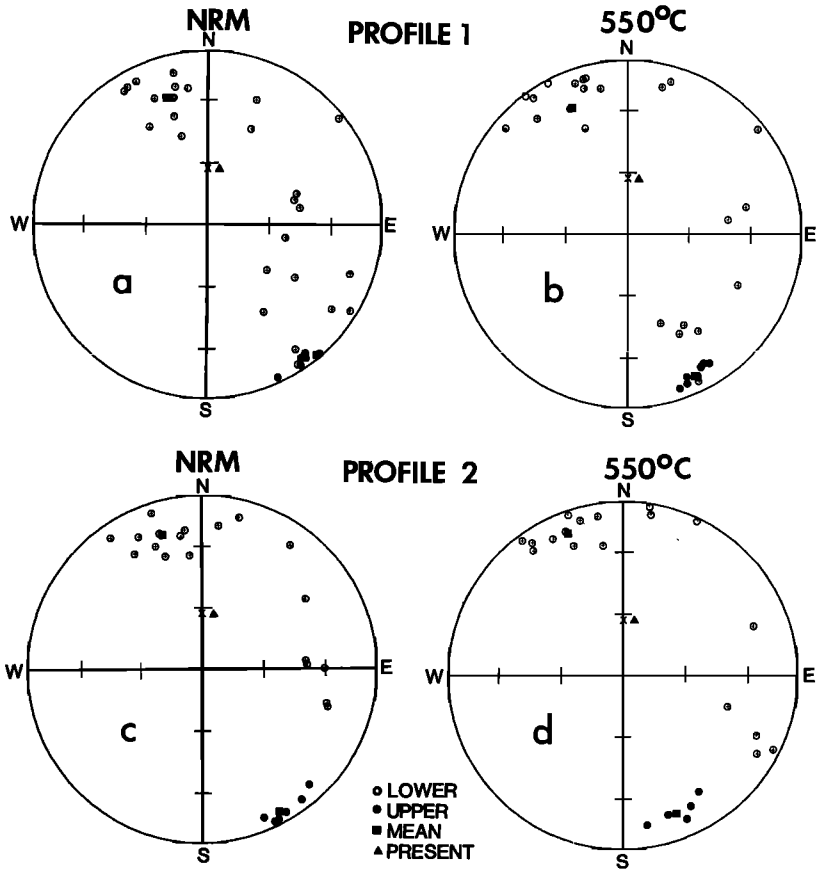


Fig. 2. Equal area plots of the NRM and 550°C directions of profiles 1 and 2. Crosses indicate the axial dipolar field; solid triangles represent the present earth's field direction.

ples obtained by *Helsley* [1969] (normal samples); they show an entirely different magnetic behavior from pilot samples reported by *Lienert and Helsley* [1980], which show regular decreases in their intensities up to 650°C. Moreover, *Lienert and Helsley* [1980] observed three samples that define a change in magnetization characteristics close to 550°C, which may indicate the presence of magnetite (Curie point 578°C). Such evidence for magnetite has not been observed in Chugwater samples. The intensity of these specimens (normal and reversed) falls off rapidly at temperatures of greater than 660°C, indicating a Curie temperature close to that of hematite (675°C) (see Figures 4*b* and 5).

The demagnetization curves presented in Figure 4*b* show an apparently uniform behavior; all of them are characterized by thermally discrete components [*Irving and Opdyke*, 1965], especially specimens 93, 126, and 127, in which the intensity remains unchanged up to temperatures near the Curie point. These characteristics agree with the results reported by *Collinson* [1974, Figure 2] and *Elston and Purucker* [1979] in red beds of the Moenkopi Formation, where the progressive thermal and chemical demagnetization analyses indicated that the most stable magnetization resides in specularite. Similar behavior is seen in samples of the Moenkopi Formation at Bears Ears Utah [*Lienert and Helsley*, 1980] in which a group of pilot samples shows a slight decrease in intensity up to 200°C and then remains fairly constant up to temperatures of 650°C, where the intensities fall off rapidly, indicating a Curie temperature close to that of hematite.

The directional changes observed in four of the samples are shown in stereographic projection in Figure 4*a* and the Zijderveld plots of two of these are shown in Figure 5. In these two cases the samples show a stable normal direction at temperatures exceeding 550°C; this is supported by the behavior of the point representing temperatures above 550°C that trend toward the origin of the Zijderveld diagram [*Zijderveld*, 1967]. This suggests that the NRM consists of two components: a very small secondary component (viscous?) and a dominant component that could be compared favorably with the expected Triassic direction for North America [*Helsley and Steiner*, 1974; *Van der Voo and Grubbs*, 1977]. The predominant component is the only component seen above 620°C as shown in Figure 5. Transitional samples 125 and 133 are essentially univectorial, as are the normal and reversed samples 93 and 141. This univectorial behavior strongly suggests that the transition samples consist of a single component of magnetization.

On the basis of the results of the demagnetization experiments with individual samples, the rest of the specimens were demagnetized at temperatures of 550°C (profiles 1 and 2), and profile 3 results at 500°C, 620°C, and 660°C are shown in Figures 2, 3, and 6–9.

Many samples could not be thermally demagnetized to 500°C, for a number of the specimens had broken during drilling in the field, and it was necessary to glue them together with epoxy, which breaks down at relatively low temperatures (see difference in samples populations in Table 2).

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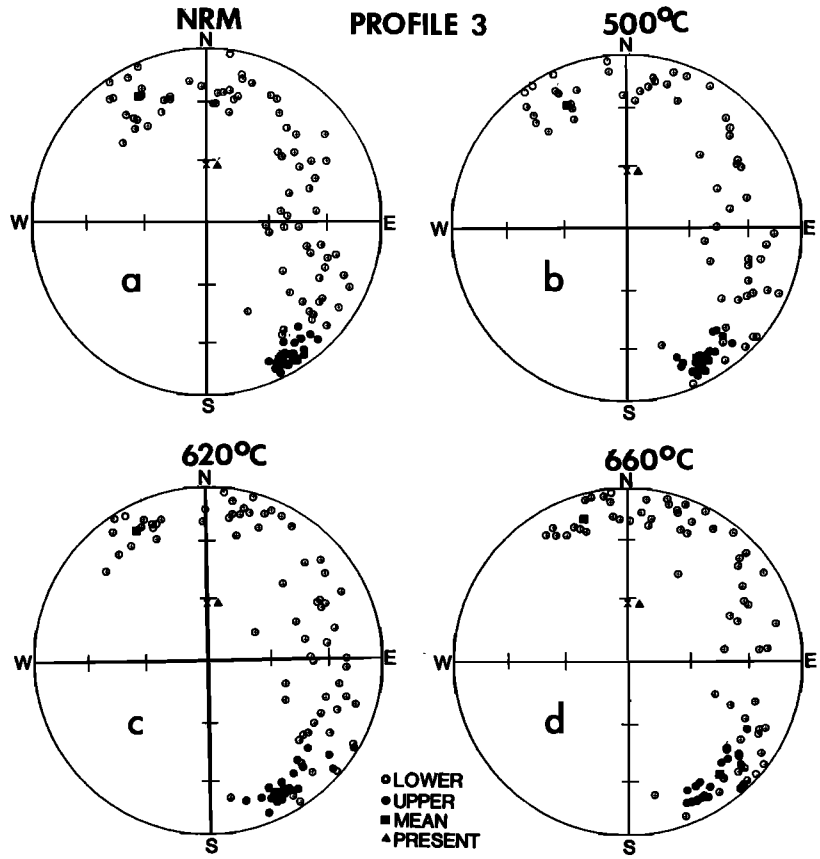


Fig. 3. Equal area plots of the NRM, 500°C, 620°C, and 660°C directions of profile 3. Crosses indicate the axial dipolar field; solid triangles represent the present earth's field direction.

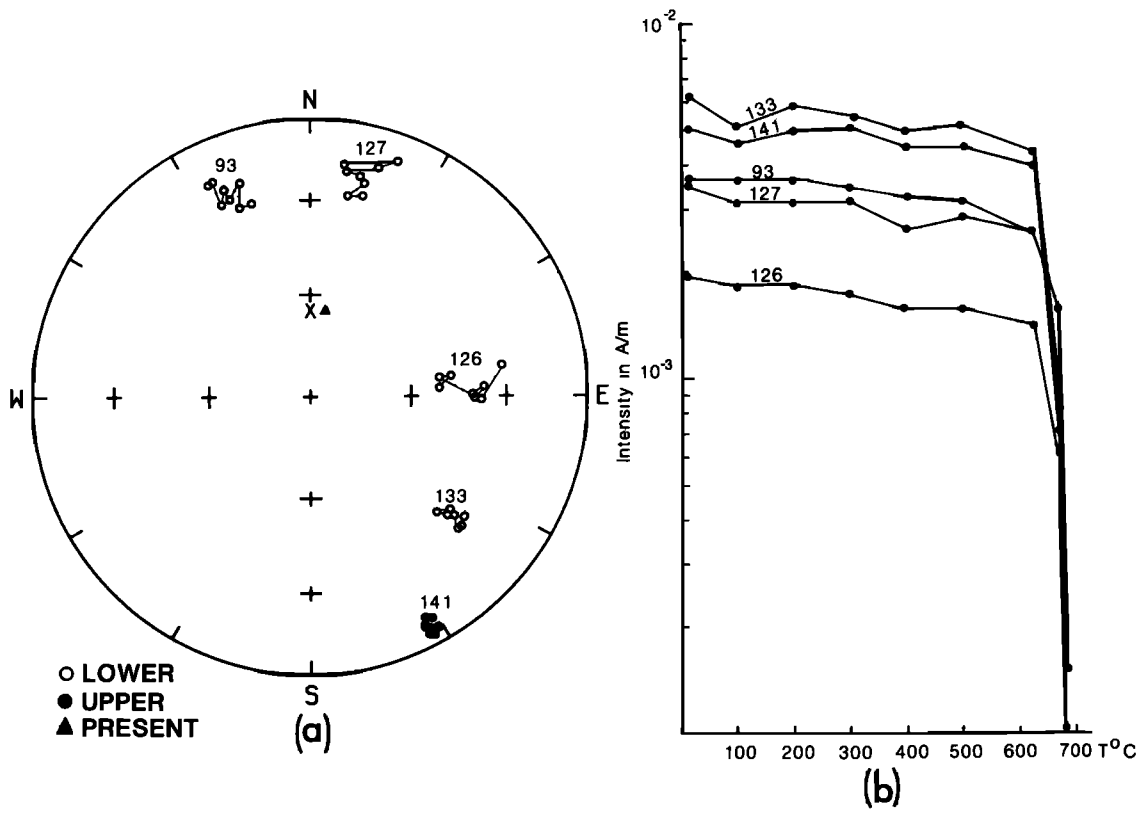


Fig. 4. Response of selected samples to progressive thermal demagnetization: (*a*) equal area plot of changes in direction; (*b*) changes in intensity. Arrows indicate the direction of change with increasing temperature. Cross indicates the axial dipolar field direction; solid triangle represents the present earth's field direction.

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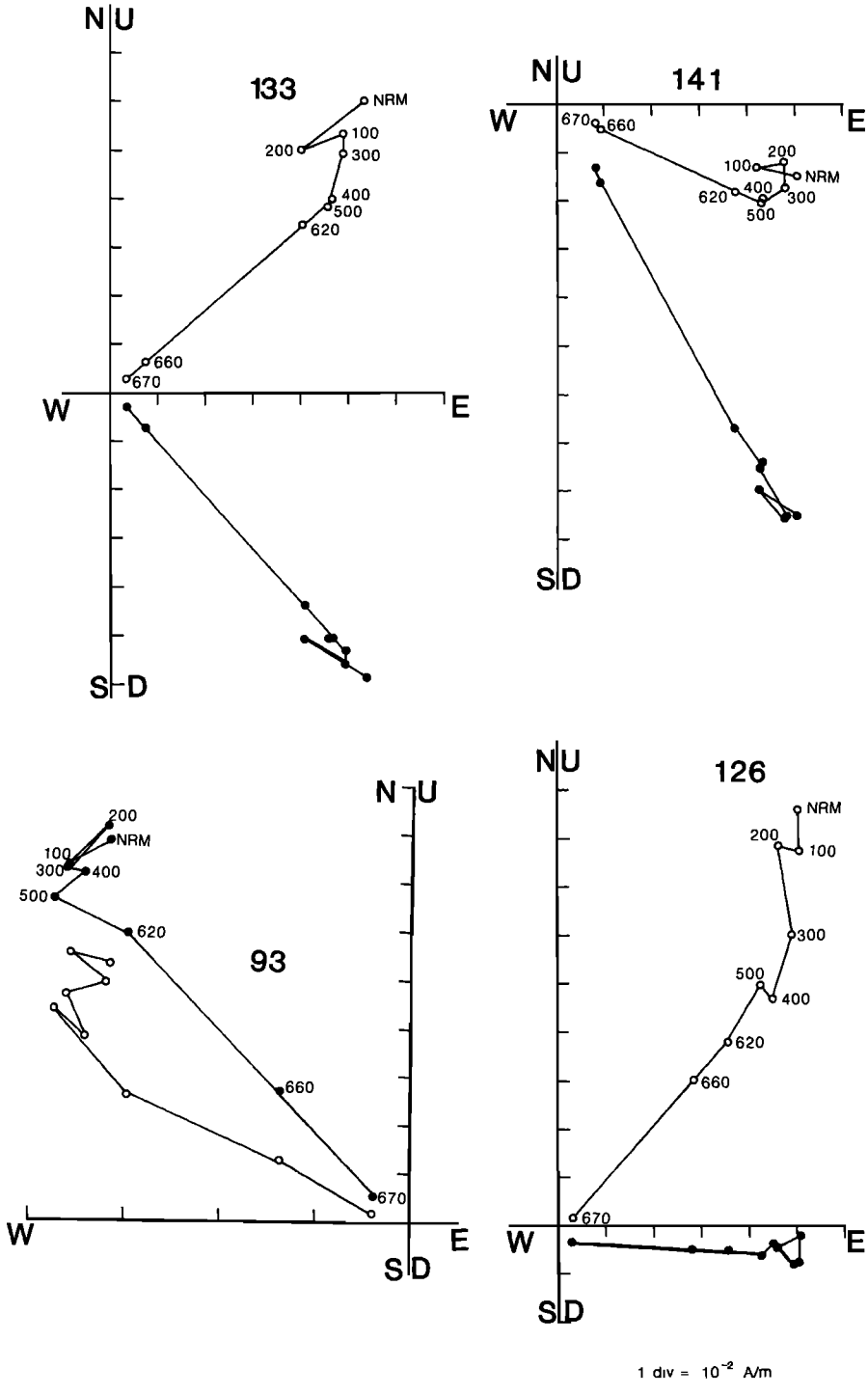


Fig. 5. Diagrams of progressive thermal demagnetization of samples 133, 141, 93, and 126 of the red sediments obtained in this study. Plotted points represent successive positions—in orthogonal projection—of the end point of the magnetic vector. Solid symbols represent projections on the vertical plane. Numbers denote temperatures in degrees Centigrade. Note that samples 126 and 133 are transitional samples.

Figure 3 shows data for 500°C and 620°C, and visual examination of the data suggests that the best step of demagnetization is 620°C, although Table 2 indicates that the 500°C step yields slightly the best statistical parameters. This suggests that demagnetization in the 500°C to 600°C range is both appropriate and adequate for this sample suite. Table 2 summarizes the mean directions and statistical parameters calculated for the normal and reversed intervals from the NRM and thermal demagnetization data at 500°C, 550°C, 620°C, and 660°C. The normal interval shows a movement away from the present earth's field toward the mean Moenkopi normal direciton reported by *Helsley* [1969, Table 7] and *Lienert and Helsley* [1980, Table 1]. The reversed interval is tightly grouped and shows a discrepancy of 18° (profile 3, 620°C) from the mean declination and zero degrees in mean inclination with respect to the reversed (500°C)

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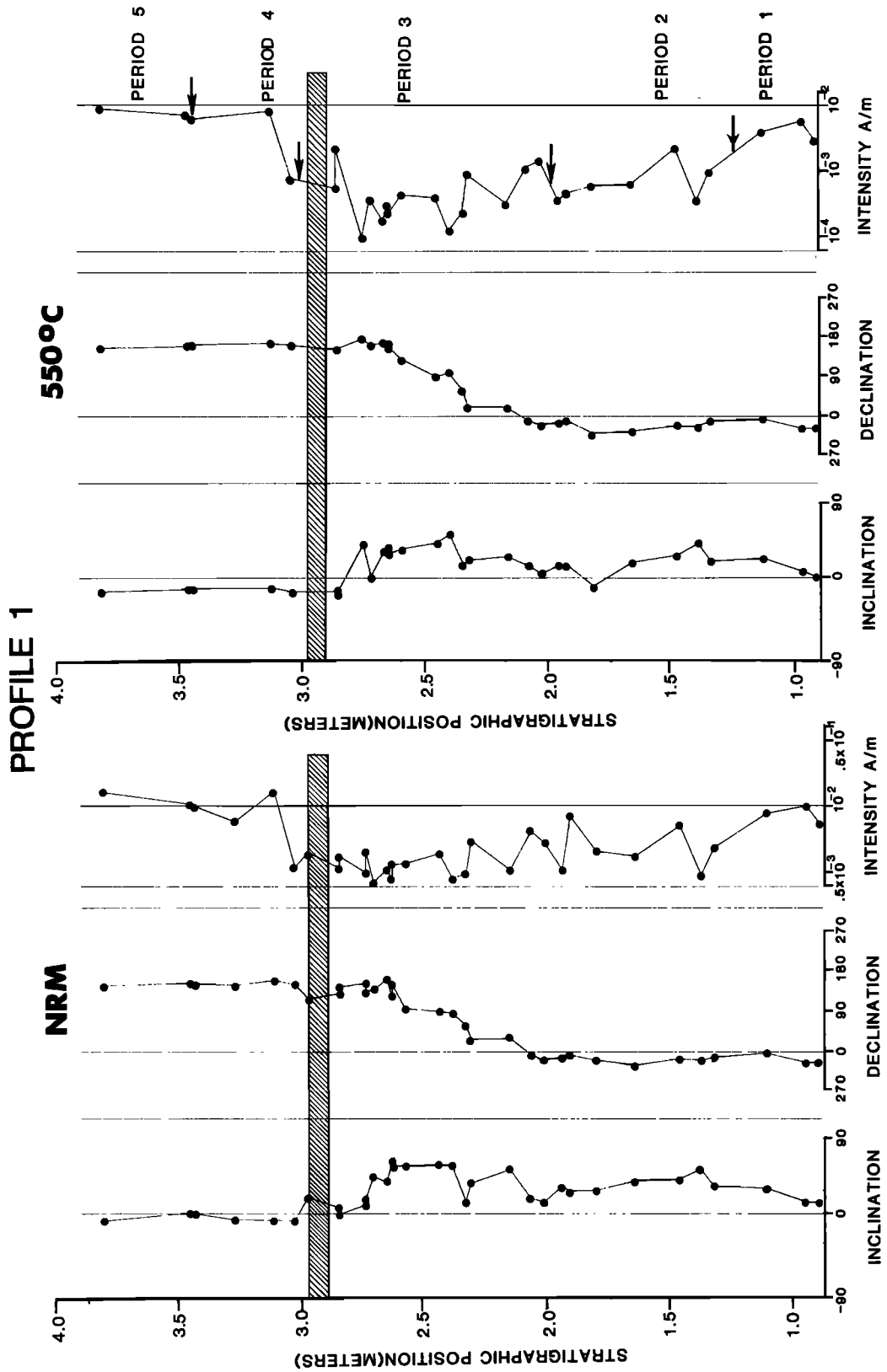


Fig. 6. Stratigraphic plot of inclinations, declinations, and intensities of the NRM and 550°C of profile 1. The hachured areas represent a marker horizon.

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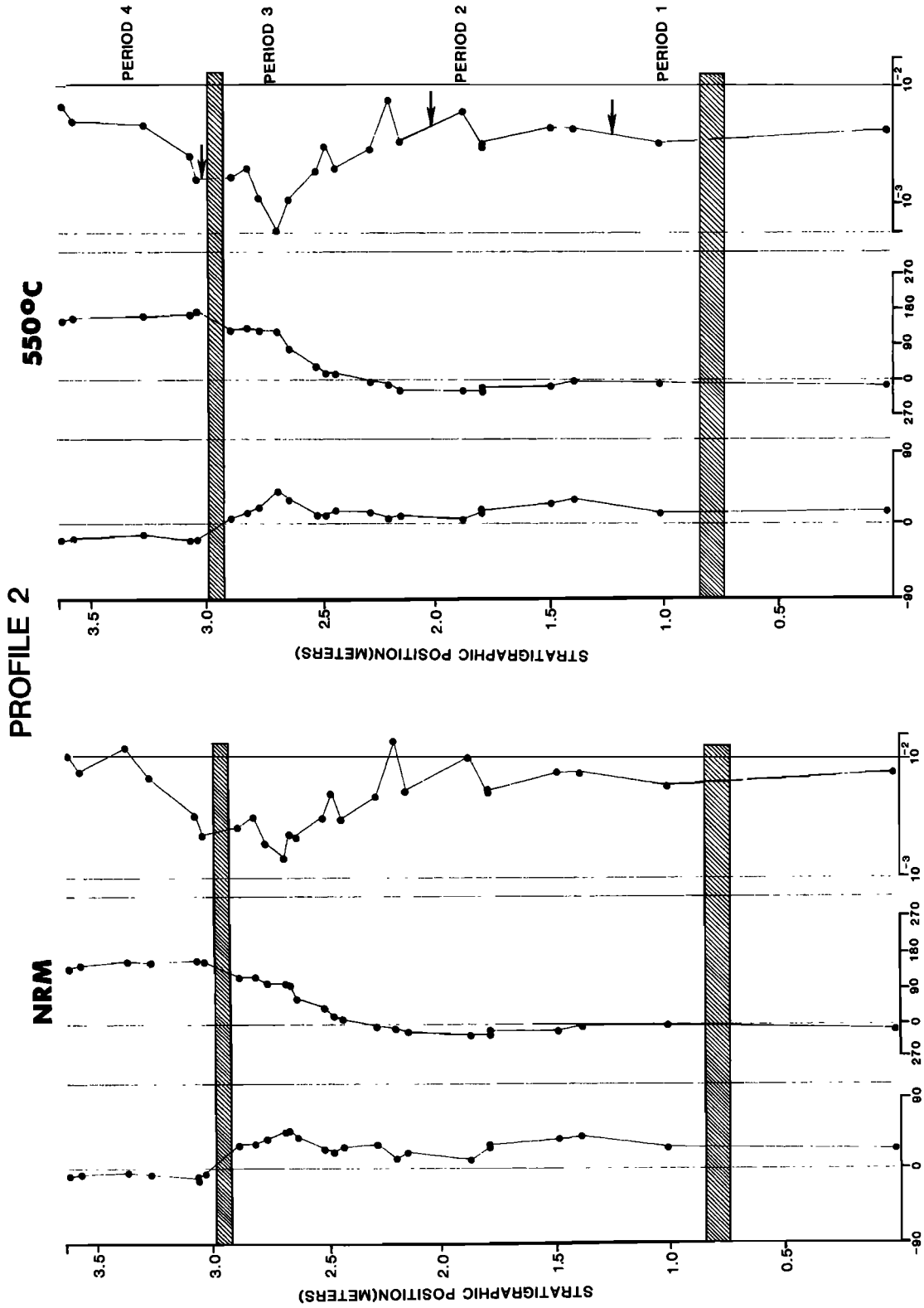


Fig. 7. Stratigraphic plot of inclinations, declinations, and intensities of NRM and 550°C of profile 2. The hachured areas represent marker horizons.

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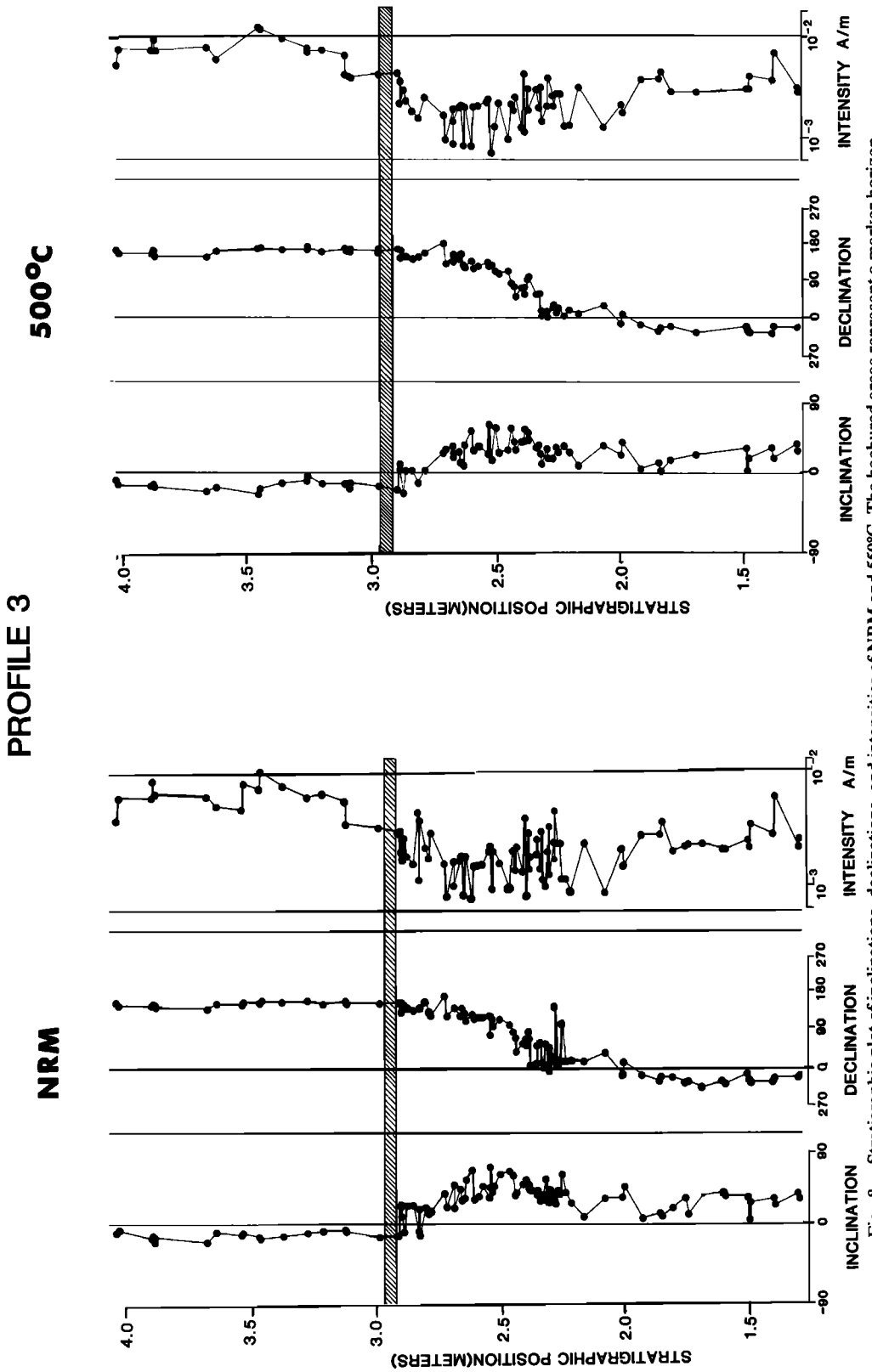


Fig. 8. Stratigraphic plot of inclinations, declinations, and intensities of NRM and 550°C. The hachured areas represent a marker horizon.

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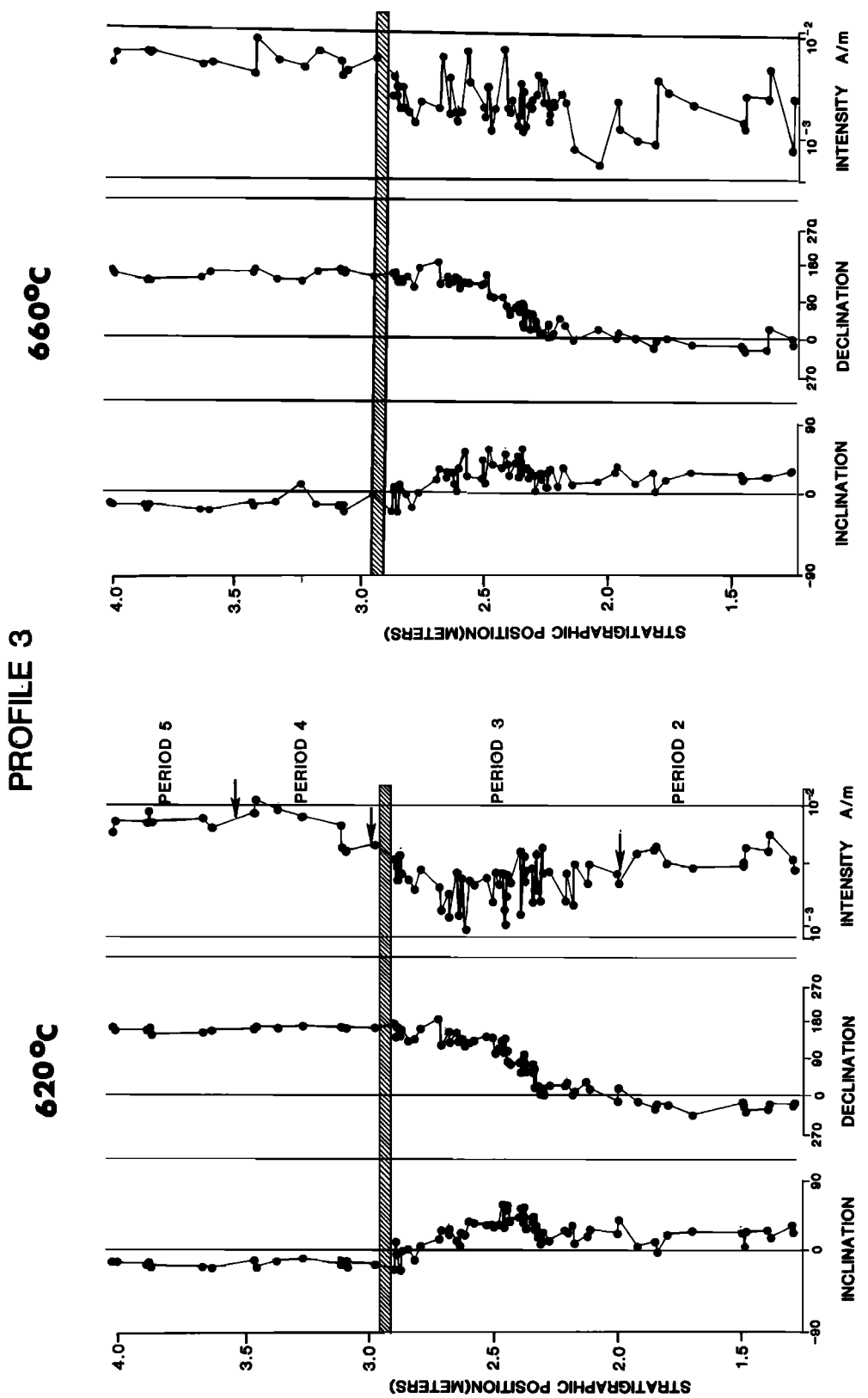


Fig. 9. Stratigraphic plot of inclinations, declinations, and intensities of 620°C and 660°C. The hachured areas represent a marker horizon.

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TABLE 1. Mean Intensity of Magnetization (1×10^{-3} A/m)

	Normal		Transition:	Reversed		
	Period 1	Period 2 Dipole Decreasing	Period 3 Nondipole Dominance	Period 4 Dipole Increasing	Period 5	All
Profile 1						
NRM	8.73	5.06	5.00	8.00	11.03	7.56
550°C	8.08	2.63	4.25	7.42	10.82	6.64
550/NRM	0.93	0.52	0.84	0.93	0.98	0.84
Profile 2						
NRM	7.32	8.17	2.96	7.88	...	6.58
550°C	5.86	7.18	2.16	7.29	...	5.62
550/NRM	0.80	0.88	0.73	0.93	...	0.84
Profile 3						
NRM	...	3.07	2.85	6.84	8.44	5.29
620°C	...	2.48	1.94	4.01	7.53	3.99
620/NRM	...	0.81	0.68	0.59	0.89	0.75

selected mean reported by *Helsley* [1969, Table 7] for the Moenkopi Formation. The reason for this difference is unknown, but it should be noted that the Moenkopi samples used in Helsley's 1969 study are not demagnetized above 500°C and appear to have considerable scatter remaining after demagnetization. The only significant effect of the additional steps of demagnetization (620° and 660°C) was a small reduction in scatter in the normal interval.

At this point the most striking feature of the stereographic projections and the stratigraphic plots of the three profiles is the presence of intermediate directions of magnetization between the normal and reversed intervals that could be easily correlated over tens of meters by means of marker horizons (hatched lines in Figures 6 to 9). The intermediate directions represent a polarity transition due to (1) a definite directional change of 180° and (2) a conspicuous decrease in intensity of magnetization (in this particular case coincident with the directional change). Figures 6 to 9 show stratigraphic plots of the declination, inclination, and intensity data both before and after thermal demagnetization.

Behavior of the Geomagnetic Field During a Polarity Transition

Intensity of Magnetization of the Samples

Picard [1964] analyzed remanent intensities of undemagnetized samples from the Chugwater Formation in terms of lithologic types and in terms of normal and reversed intervals. He found that the distributions within normal and reversed groups were identical and that the only differences present were between lithologic types. His mean intensity for all samples was 6.5×10^{-3} A/m, and his range was from 0.6 to 70.4×10^{-3} A/m.

Within the profiles the mean intensity of all samples is 5.33×10^{-3} A/m and the range of NRM intensities is from 4.04 to 246×10^{-3} A/m. Figures 6 to 9 show stratigraphic plots of the entire sections. Careful examination of these data, their running averages, and the mean intensities in Table 1 suggests that the intensity of the magnetic field can be subdivided into five intensity periods. These intensity periods, identified by either apparent changes in intensity or direction, are somewhat subjective but nevertheless form a basis for model discussions. From the base of the sections up, these periods are as follows: period 1—typical normal field intensity from 0 to approximately 1.20 m; period 2—generally a time of decreasing intensity from 1.20 m to approximately 2.0 m; period 3—a period of very low field intensity from 2.0 m to 3.0 m; period 4—a period of increasing field intensity from 3.0 m to approximately 3.5 m; and period 5—a period of typical reversed field intensity from 3.5 m to approximately 4.0 m (top of profile 3). (See Figures 6 through 9.)

There is a slight difference in the mean intensity of magnetization between the normal and reversed groups, but there is a remarkable difference between both the NRM and thermally demagnetized mean intensities of the reversed and normal groups and the mean intensity in the transition zone (see Table 1). Two features are immediately apparent. First, there is a pronounced decrease in the intensity of magnetization upon demagnetization of the samples in the transition zone with respect to the normal and reversed intervals (Figures 6 to 9). Second, the values of mean intensities for the three profiles after heating to 550°C and 620°C are lower with respect to the mean intensities of their respective NRM data (see Table 1). As mentioned above, the transition zone is divided into five periods depending upon the intensity and direction of magnetization of the samples. The true dipolar field is represented by periods 1 and 5. The lower intensity in periods 2 and 4 in the transition zone probably reflects a decrease in the intensity of the earth's magnetic field without any significant change in direction and thus supports the idea of a dipolar field reducing to a minimum value close to zero followed by an increase in the opposite sense, following the transition. Period 3 coincides with the onset of the directional change, and its maximum intensity decrease is coincident with the steepest inclination (Figures 6 to 9). This is consistent with a model in which the dipolar field has decreased to near zero and the field then becomes dominated by the nondipolar portion. Low intensities of magnetization have been generally noted in polarity transitions [*Lawley*, 1970; *Dagley and Lawley*, 1974; *Baag and Helsley*, 1974; *Hillhouse and Cox*, 1976], and thus it is worth noting once again that within period 3 the intensity of magnetization drops to a value that is about 25% of the average intensity observed after the dipolar field has recovered; this is in close agreement with the 20% of the nondipole component reported by *Cox et al.* [1975], and thus it might be correct to infer that period 3 is a period of nondipole dominance. On the other hand, if instead of taking the mean value of the nondipole dominant period one takes the mean of the minimum values of magnetization (at least three samples per profile) in the

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transition zone and compares this value with the mean value of the reversed field after demagnetization to 550°C and 660°C, it represents about 9% of the dipole field in Early Triassic time. This result agrees with the result given by *Irving* [1964], who reports that the rms strength of the nondipole field at the surface amounts of about 5% of today's dipole field. Although this does not prove that period 3 is a nondipole dominant period, it is strongly suggestive.

Duration of the Reversal and Timing of Phenomena Within the Reversal

The data set indicates that a slow change in intensity, without change in direction, begins long before the marked intensity decrease associated with the change in direction. The observed intensity decrease accompanying the intermediate directions in these profiles is shown in Figures 6 to 9. The major decrease in intensity of magnetization appears to be coincident with the directional change and recovers shortly after the directional change is complete. The present results appear to indicate that the major portion of the intensity change took the same amount of time as the directional change. The less pronounced intensity lows of periods 2 and 4, however, precede and follow the directional change and suggest that the total time involved in the transition may be three times more than that associated with the period of rapid directional change.

Based on profiles 1 through 3, the average thickness of the transition zone is 1.0 m. From this one can estimate the duration of the directional change and intensity excursion. Using the same type of argument as that presented by *Picard* [1964], namely, thickness divided by an estimate of the number of years represented by the formation, the estimated average depositional rate for the formation is about 41,000 y/m of sedimentary rock. However, it is unlikely that any one part of the formation is deposited at this extremely slow rate. If one assumes that most of the time is represented by hiatuses, i.e., periods of nondeposition or erosion, then the well-bedded upper portion of the formation should represent more time while the poorly bedded lower part of the formation (Red Peak Member sampled for this study) would represent a considerably higher than normal sedimentary rate. Thus the duration of the transition phenomenon would have to have taken place in substantially less than 36,900 years. How much less, unfortunately, cannot be determined. Without any better information on rates of sedimentation of parts of the Chugwater Formation, no conclusion can be made about the length of time represented by the 1.0 m of sediments in which the intermediate directions are recorded.

From speculation based on average rates of shelf sedimentation (*Picard* [1966] considers the 'alternating facies' to be an alternating of paralic and shallow marine environments), a sedimentation rate of about 15 to 700 mm/1000 years [*Fischer*, 1969] can be assumed. Under this assumption the polarity transition might have taken as long as 27,000 years or as short as 600 years. Thus the range of estimates is compatible with the estimates made by others but in no way contributes any additional understanding of the time required for transitions.

Comparing the transition pattern (declination and intensity) of this study with the transition recorded in the Moenkopi Formation reported by *Baag and Helsley* [1974], the observed data from the Chugwater Formation fit the model of a rapid acquisition pattern that is due to a rapid acquisition component only. This rapid acquisition pattern can be exhibited by a detrital remanent magnetization (DRM) and also by any rapid chemical magnetization, either with or without a certain time lag.

The results obtained in this study indicate that the duration of the intensity change (period 3) is the same as the duration of the directional change and that the polarity transition takes place in approximately 1 m of sediment. These results are in agreement with the results of transition studies in the red bed sequences reported by *Khramov* [1960], *Picard* [1964], *Petrova* [1965], *Helsley* [1969], and *Baag and Helsley* [1974], as well as other studies of sedimentary rocks [*McElhinny*, 1970; *Pullaiah and Verma*, 1970]. Moreover, the intensity patterns for the transition interval recorded by these red beds are also compatible with those recorded by deep-sea sediments, lava flows, and intrusive rocks, such as the results of *Ninkovich et al.* [1966], *Harrison and Somayajulu* [1966], *Dunn et al.* [1971], *York et al.* [1971], *Steinhauser and Vincenz* [1973], *Opdyke et al.* [1973], *Hillhouse and Cox* [1976], *Dodson et al.* [1978], and *Hammond et al.* [1979]. The results support the hypothesis that the time duration of field intensity decrease is the same as that of direction changes and consequently indicate that the dipole field 'collapsed' rather than 'flipped over' during the reversal, assuming that the low intensities of the demagnetized data during the transition reflect a decrease in the intensity of the geomagnetic field [*Hillhouse and Cox*, 1976].

Virtual Geomagnetic Poles

As an alternative presentation of the data, the results are plotted as a succession of virtual geomagnetic poles (VGP) defined as the position of the equivalent geomagnetic pole calculated from a spot reading of the paleomagnetic field. They represent only an instant in time, just as the present geomagnetic poles are instantaneous [*McElhinny*, 1973]. It should be noted that this is not intended to imply that the field was dipolar during the reversal. As shown by *Dodson et al.* [1978], VGP's afford a convenient presentation, particularly if comparisons between different records are to be attempted.

The reversal is best defined in profile 3 and is defined by 83 VGP's (48 are intermediate). The record certainly falls in the 'A' category of *Fuller et al.* [1979]. The VGP's make a fairly smooth transit from normal to reversed polarity, with the path confined to a sector of longitude between 85°W and 110°E. Following the reversal's onset, the virtual pole moved slowly across northern latitudes, then lingered briefly between 60°N and 40°N and between 35°E and 70°E, then again between 20°N and 40°N and from 335°E to 5°E, suggesting the presence of brief periods of stability of the field (or nondipolar field dominance) or fluctuations in sedimentation rate. Before completing its transit to reversed polarity, the VGP's representing the transitional field seemed to recover very slowly in comparison with the behavior of the VGP's representing the field right after the reversal's onset (see Figure 10).

The systematic and reproducible transition VGP path determined in this Chugwater reversal may be the result of a sluggish paleomagnetic recorder having smoothed a more rapidly changing magnetic field, each horizon representing an integration of the field over a period of several years or several centuries. The sedimentation rate may not have been rapid enough and the remanence acquisition time not short

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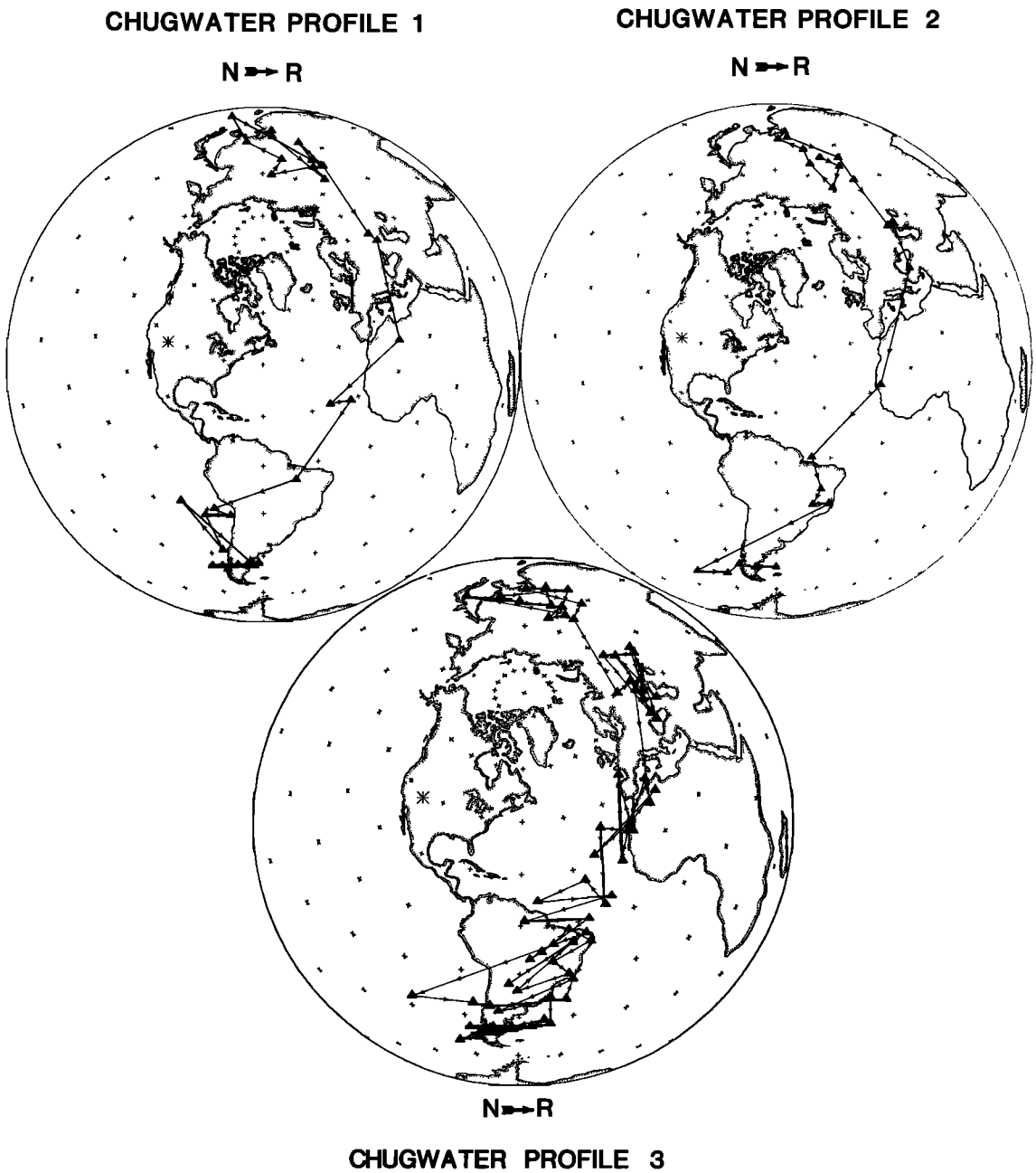


Fig. 10. Path of virtual geomagnetic poles for a normal to reversed polarity transition observed in the Chugwater Formation, Wyoming. Profiles 1, 2, and 3.

enough to permit full resolution of the transition pole path. A hint of this 'swinging' pattern is present if a sample by sample analysis is made. A similar result has been reported by *Hillhouse and Cox* [1976] on Quaternary sediments at Lake Tecopa, California. These relatively smooth pole paths can be contrasted with other well-documented transitions, for example, the sequence of 26 lava flows spanning an Upper Miocene polarity transition reported by *Watkins* [1969], where he found large swings in magnetic direction between successive flows, or the results reported by *Dunn et al.* [1971] within a Miocene granodiorite intrusion which also showed rapid swings. The above cited results on lavas suggest that lavas are instantaneous recorders of the geomagnetic field and these rapid swings are probably real and should probably be expected in any set of data. Thus one is left with the conclusion that the lack of fluctuation may indicate that the field has been imperfectly recorded and that the time of magnetization was not 'instantaneous' but simply rapid or penecontemporaneous, as suggested by *Baag and Helsley* [1974].

Interpretation of Transitional Fields in Terms of Axisymmetric Fields

Assuming that the behavior of a transitional field is nondipolar, one could compare the behavior of this polarity transition with the polarity transitions summarized by *Fuller et al.* [1979] in order to see if this transition fits the transitional field configuration models proposed by *Hoffman and Fuller* [1978]. Analysis of polarity transition records, through the determined transitional path of the VGP's, has

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TABLE 2. Statistical Analyses of the Normal and Reversed Intervals

Profile	Latitude, °N	Longitude, °W	Demag., °C	N	D	I	K	α_{95}	Pole Longitude, °E	Pole Latitude	dp	dm
1 R	43–40	109–40	NRM	9	140	1	27.1	10.1	301.0	33.8S	5.0	10.1
1 N	43–40	109–40	NRM	12	342	23	34.4	7.5	103.2	54.8	4.2	8.0
Mean			NRM	21	332	13	15.1	8.5	112.4	45.9	4.4	8.6
1 R	43–40	109–40	550	7	154	−11	204.5	4.2	289.4	45.6S	2.2	4.3
1 N	43–40	109–40	550	12	337	23	34.4	7.5	103.2	54.8	4.2	8.0
Mean			550	18	336	12	52.8	4.8	107.7	46.9	2.5	4.9
2 R	43–40	109–40	NRM	6	151	−6	101.4	6.7	291.3	41.8S	3.4	6.7
2 N	43–40	109–40	NRM	10	344	22	41.4	7.6	99.1	54.9	4.2	8.0
Mean			NRM	16	339	16	30.0	6.8	104.6	50.0	3.6	7.0
2 R	43–40	109–40	550	5	159	−16	73.8	8.9	284.0	50.0S	4.7	9.2
2 N	43–40	109–40	550	9	338	12	45.7	7.7	104.9	47.9	3.9	7.8
Mean			550	14	338	14	55.3	5.4	104.7	48.7	2.8	5.5
3 R	43–40	109–40	NRM	19	151	−10	166.3	2.6	292.9	43.5S	1.3	2.6
3 N	43–40	109–40	NRM	16	333	21	34.0	6.4	114.9	49.9	3.6	6.8
Mean			NRM	35	332	15	46.3	3.6	113.7	46.4	1.9	3.7
3 R	43–40	109–40	500	22	152	−13	94.0	3.2	292.6	45.6S	1.7	3.3
3 N	43–40	109–40	500	11	335	22	35.4	7.8	111.9	51.1	4.3	8.2
Mean			500	33	333	16	53.5	3.5	112.4	47.4	1.8	3.5
3 R	43–40	109–40	620	16	149	−13	48.3	5.4	296.2	44.3S	2.8	5.5
3 N	43–40	109–40	620	11	332	16	43.4	7.0	113.9	47.0	3.7	7.2
Mean			620	27	330	14	46.8	4.1	115.3	45.4	2.2	4.2
3 R	43–40	109–40	660	14	141	−16	41.6	6.2	306.0	40.8S	3.3	6.4
3 N	43–40	109–40	660	11	343	17	47.9	6.7	99.0	52.0	3.6	6.9
Mean			660	25	331	16	26.0	5.8	115.5	46.5	3.1	6.0
1, 2, 3 R	43–40	109–40	NRM	3	150	−5	92.5	12.9	293.2	40.8S	6.5	12.9
1, 2, 3 N	43–40	109–40	NRM	3	342	22	214.6	8.4	102.8	54.2	4.7	8.9
Mean (NRM)			NRM	6	335	14	38.7	10.9	108.8	47.5	5.7	11.2
1, 2, 3 R	43–40	109–40	550	3	157	−13	353.9	3.5	286.4	48.0S	3.4	6.7
1, 3, 3 N	43–40	109–40	550	3	338	19	168.1	9.5	106.1	51.5	5.2	9.9
Mean (550°)			550	6	338	16	196.4	4.8	106.2	49.7	2.5	4.9
3 R	43–40	109–40	620	16	149	−13	48.3	5.4	296.2	44.3S	2.8	5.5
3 N	43–40	109–40	620	11	332	16	43.4	7.0	113.9	47.0	3.7	7.2
Mean (620°)			620	27	330	14	46.8	4.1	115.3	45.4	2.2	4.2
3 R	43–40	109–40	660	14	141	−16	41.6	6.2	306.0	40.8S	3.3	6.4
3 N	43–40	109–40	660	11	343	17	47.9	6.7	99.0	52.0	3.6	6.9
Mean (660°)			660	25	331	16	26.0	5.8	115.5	46.5	3.1	6.0

N, number of samples; D, declination in degrees, east of north; I, inclination in degrees, positive downward; K, Fisher's precision parameter; α_{95}, semiangle of cone of 95% confidence for mean directions; and (dp, dm), oval of 95% confidence about the pole position.

shown certain characteristics to be common to many transitions. It is of particular interest that most detailed transitional VGP paths are largely constrained in longitude during the excursion from one polarity to the other. Two such paths from widely separated sites which recorded the Matuyama-Brunhes reverse-to-normal (R → N) transition were found to be widely dissimilar, thus providing clear evidence that the ambient field during this reversal was predominantly nondipolar [*Hillhouse and Cox*, 1976]. Further clarification of the geometry of transitional fields has been made recently with the recognition of a dependence of the transitional VGP path behavior on site location [*Hoffman*, 1977]. Moreover, this site control has been shown to be consistent with a transitional field which is predominantly zonal, a field geometry suggestive of a hydromagnetic reversal process which starts at certain latitudes in the core [*Hoffman*, 1977].

Both octupole [*Hoffman*, 1977] and quadrupole [*Dodson et al.*, 1978] dominated transitional field geometries have been hypothesized by *Hoffman and Fuller* [1978]. Furthermore, they have shown that these field configurations and, hence, primary characteristics of the reversing dynamo process from which each may arise are paleomagnetically distinguishable. They also indicated more specifically which cases are possible, although the lack of available data precludes any firm conclusion.

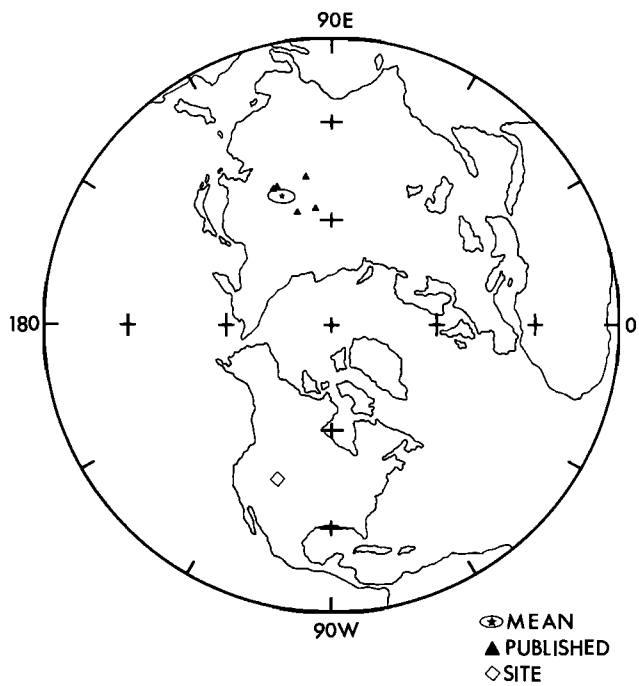


Fig. 11. Summary of Triassic poles for North America, including the mean pole for a section sampled in the Red Peak Member in the Chugwater Formation.

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TABLE 3. Summary of Lower Triassic Poles

Formation	Latitude, °N	Longitude, °E	D	I	K	α_{95}	Reference
Chugwater (Lander)	47	100	344	27	55	4	*Collinson and Runcorn* [1960]
Chugwater	48	112	344	17	51	5	*Irving* [1964]
Chugwater	48.5	112.2	154	−15.5	86	5.9	*Van der Voo and Grubbs* [1977]
Moenkopi	57	107	338	19	21	13	*Collinson and Runcorn* [1960]
Moenkopi	57	98	346	17	85	5	*Helsley* [1969]
Chugwater	45.4	115.3	330	27	47	4.1	this study

At this point it is worth noting that several major features of this transition are similar to those summarized by *Fuller et al.* [1979]. These features are as follows:

1. The reversal is accompanied by a decrease in intensity of field.
2. The reversal in direction is accomplished by a VGP path, which is confined in longitude; however, the mean path from pole to pole is accompanied by a certain degree of looping (possibly a period of stability or nondipolar field dominance) about certain points, so that the path is traversed somewhat irregularly.
3. The VGP path observed at a northern hemisphere site is about 90° from the sampling site; that is, the transitional poles straddle the near-side far-side boundary if one plots up the VGP's relative to the sampling site.
4. The decrease in intensity seems to be coincident with the time taken for the change in direction, and the duration of the reversal is comparable in length to that reported by other studies (e.g., 10,000 years proposed by *Cox et al.* [1975]).

PALEOMAGNETIC POLES

Pole Positions

The pole positions derived from both the NRM and thermal demagnetization measurements presented above in Figures 6 through 9 are given in Table 2, which includes the means of the normal and reversed intervals, as well as the mean of both intervals.

The best statistical parameters were obtained for the various profiles at a demagnetization step between 500°C and 620°C. In the case of profile 3 we have used 620°C rather than 500°C (the best statistics) because the normal and reversed data are more nearly antipodal. Thus the selected demagnetized data from both the reversed and normal intervals are considered to be the best estimates of the pole positions for the Red Peak Member of the Chugwater Formation near Dubois, Wyoming. The pole has a cone of confidence at the 95% level [*Fisher*, 1953] of about 4.1°, and the precision of the pole as reflected in the oval of confidence (*dp, dm*) is quite acceptable for both intervals (*dp* =

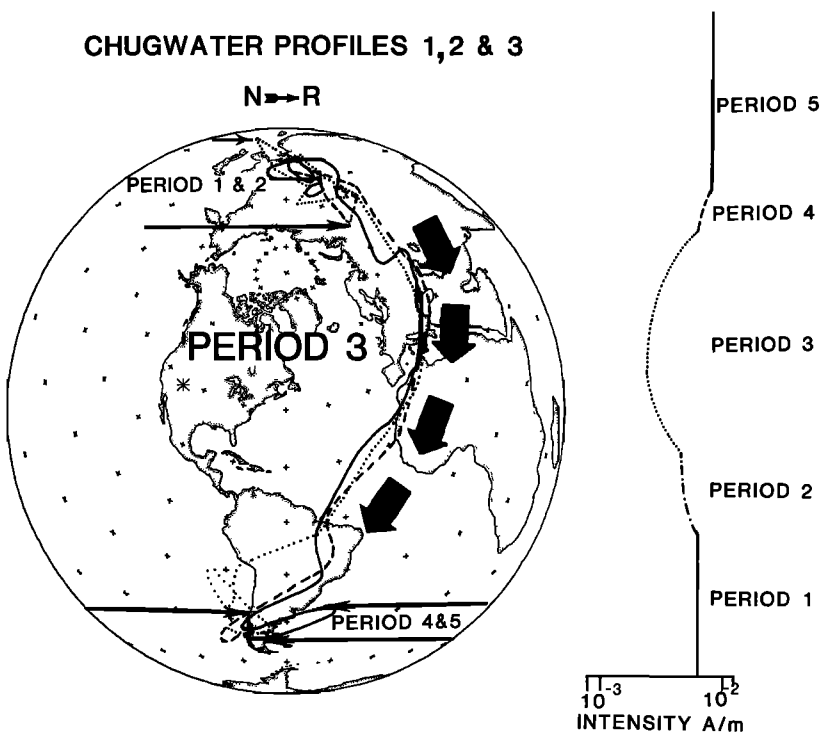


Fig. 12. Correlation between the N → R VGP average paths of profiles 1, 2, and 3 and a model intensity of magnetization of the Chugwater Formation.

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2.2° and dm = 4.2°); the normal and reversed polarities are about 175° apart, indicating that a stable magnetization has been isolated. The VGP of the normal and reversed groups combined is given by the star in Figure 11.

Comparison With Triassic Paleomagnetic Poles for North America

A summary of the Triassic paleomagnetic pole positions for North America is given in Figure 11 and Table 3. The pole determined in this study lies approximately in the center of the main group of poles reported for other Triassic units (see Table 3 for references). The results reported by *Collinson and Runcorn* [1960] are based on samples not treated thermally and thus should not be given the same reliability as those for the more recent studies in which demagnetization studies have been made.

SUMMARY AND CONCLUSIONS

Stable directions of magnetization of the samples taken for this polarity transition study show once again that red sediments are good recorders of geomagnetic phenomena. The clear intermediate directions of magnetization after demagnetizing the specimens to 550°C and 660°C represent a true polarity transition and not two opposed magnetizations of compatible intensity superimposed on each other. This conclusion is supported by the correlation of intermediate directions of magnetization of the same reversal at three different sampling localities that gave the same paleomagnetic answer (identical magnetic behavior upon demagnetization, and VGP paths) displaying a remarkable decrease in intensity coincident with a 180° directional change during the polarity transition. The intensity of magnetization of the section can be subdivided into five intensity periods, two dipolar periods, one dipolar decreasing period, a nondipolar dominant period, and a dipolar increasing period (Figure 12). The nondipolar period took place in about 1.0 m of red sediments, and the mean intensity of magnetization drops to a value that is about 25% of the mean intensity observed after the dipolar field has recovered; or if one compares the values of the mean of the least magnetized samples with the mean value of the reversed field after demagnetization to 550° and 620°C, it represents about 9% of the dipole field in the Early Triassic. These conclusions lead us to think that the geomagnetic field during this Chugwater polarity transition was nondipolar and support the hypothesis that the time duration of field intensity decrease was longer than that of direction changes and consequently indicate that the dipole field 'collapsed' rather than flipped over during the reversal.

The VGP path, confined to a 20° wide sector of longitude between approximately 90° from the site (Figure 10), shows a fairly smooth transit from a normal to a reversed polarity. Following the onset of reversal, the VGP moved slowly across northern latitudes, then lingered briefly between 60° and 40°N (between 35° and 70°E), then again between 20° and 40°N (between 335° and 5°E), suggesting the presence of brief periods of stability of the field (or nondipolar field dominance). Before completing the transit to the reversed polarity, the field appears to recover more slowly if one compares the behavior of the field during recovery with that during the reversal's onset. This behavior of the geomagnetic field can be correlated with the nondipolar dominance period and the dipolar decreasing and increasing field depicted by the intensity plot shown in Figure 12. The VGP path shows that it is located about 90° from the site, straddling the boundary between the near side and the far side, and thus does not assist in constraining the hypothesis of *Hoffman and Fuller* [1978].

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